FIVE EQUIVALENT CHARACTERISATIONS OF A PARASITIC ADEQUATE MODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The base change formula for adequate modules and their locality. Same chapter, Lemmas lemma-adequate-flat and lemma-adequate-local, invoked at proof lines 2280-2283. Turn F(U_i) = 0 into F(V_j) = 0 for each flat V_j -> U_i.
- The comparison of the vanishing of vF with tav-parasiticity. Same chapter, Remark remark-compare, invoked at proof lines 2290-2292. Supplies the equivalence of (1) with (3).

Source reference. Original lemma 06UW: sources/stacks-adequate/pinned-original-source.tex, source lines 442–442. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06VI: sources/stacks-adequate/pinned-original-source.tex, source lines 1707–1707. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 06VN: sources/stacks-adequate/pinned-original-source.tex, source lines 1955–1955. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{F}$ be an adequate $\mathcal{O}$-module on $(Sch/S)_\tau$. The following are equivalent:*

(1) $v\mathcal{F} = 0$,
(2) *$\mathcal{F}$ is parasitic,*
(3) *$\mathcal{F}$ is parasitic for the τ-topology,*
(4) *$\mathcal{F}(U) = 0$ for all $U \subset S$ open, and*
(5) *there exists an affine open covering $S = \bigcup U_i$ such that $\mathcal{F}(U_i) = 0$ for all i.*

Date: stacks-project:adequate.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. The implications (2) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (5) are immediate from the definitions. Assume (5). Suppose that $S = \bigcup U_i$ is an affine open covering such that $\mathcal{F}(U_i) = 0$ for all i. Let $V \to S$ be a flat morphism. There exists an affine open covering $V = \bigcup V_j$ such that each V_j maps into some U_i. As the morphism $V_j \to S$ is flat, also $V_j \to U_i$ is flat. Hence the corresponding ring map $A_i = \mathcal{O}(U_i) \to \mathcal{O}(V_j) = B_j$ is flat. Thus by Lemma 06VI and Lemma 06UW we see that $\mathcal{F}(U_i) \otimes_{A_i} B_j \to \mathcal{F}(V_j)$ is an isomorphism. Hence $\mathcal{F}(V_j) = 0$. Since $\mathcal{F}$ is a sheaf for the Zariski topology we conclude that $\mathcal{F}(V) = 0$. In this way we see that (5) implies (2).

This proves the equivalence of (2), (3), (4), and (5). As (1) is equivalent to (3) (see Remark 06VN) we conclude that all five conditions are equivalent. $\square$